

FOUNDATION OF DATA SCIENCE

UNIT - 1 ASSIGNMENT

Topics: Data Science Process & Data Warehouse

Name: _____

Register No.: _____

Class / Section: _____

Subject Code: _____

Date: _____

1. DATA SCIENCE PROCESS

Introduction

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, machine learning, and domain knowledge to extract useful information and insights from data.

The Data Science Process is a systematic sequence of steps used to collect raw data, process and analyze it, build models, and communicate meaningful results for decision-making.

Steps in the Data Science Process

The major steps are: **Problem Definition -> Data Collection -> Data Cleaning -> Exploratory Data Analysis -> Feature Engineering -> Model Building -> Model Evaluation -> Deployment -> Communication & Monitoring.**

1. Problem Definition

The first step is to clearly understand and define the problem. A data scientist must identify: What problem needs to be solved? What is the objective? What data is required? What type of output is expected? How will success be measured?

Example: An online shopping company wants to predict whether a customer will purchase a product. Here, the objective is to predict customer purchase behavior using historical customer data.

2. Data Collection

After defining the problem, relevant data is collected from different sources.

Sources include databases, websites, sensors, surveys, social media, APIs, transaction records, cloud storage, and enterprise applications. Example: For predicting customer purchases, data may include customer age, gender, previous purchases, browsing history, product category, and time spent on the website.

3. Data Cleaning

Raw data usually contains errors and inconsistencies. Therefore, it must be cleaned before analysis.

Common problems include missing values, duplicate records, incorrect data, outliers, inconsistent formats, and invalid entries. Data cleaning may involve removing duplicate records, filling missing values, correcting errors, converting data types, and handling outliers. Example: If a customer's age is recorded as 250, it is probably an invalid value and needs to be corrected or removed.

4. Exploratory Data Analysis (EDA)

EDA is performed to understand the characteristics and patterns present in the data.

It involves statistical analysis, data visualization, identifying relationships, finding trends, detecting outliers, and understanding distributions. Common visualization techniques are bar charts,

histograms, pie charts, scatter plots, box plots, and line graphs. Example: A company may use a bar chart to determine which product category has the highest sales.

5. Feature Engineering

Features are the variables used by a machine-learning model. Feature engineering involves selecting, transforming, or creating useful features from existing data.

Examples: converting date of birth into age, converting transaction date into month/year, creating total purchase amount, and encoding categorical variables into numerical values. Good features can significantly improve model performance.

6. Model Building

After preparing the data, a suitable statistical or machine-learning model is selected and trained.

Depending on the problem, algorithms may include Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Naive Bayes, Support Vector Machine, and clustering algorithms. Example: A company can use Logistic Regression to predict whether a customer will purchase a product: Yes or No.

7. Model Evaluation

The developed model must be evaluated to determine how well it performs.

For classification, evaluation metrics include accuracy, precision, recall, F1-score, and confusion matrix. For regression, metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² score. The model is generally tested using data that was not used during training.

8. Deployment

Once the model performs satisfactorily, it can be deployed into a real-world application.

For example, a trained recommendation model can be integrated into an e-commerce website to recommend products to customers. Deployment may involve web applications, mobile applications, cloud platforms, business dashboards, and APIs.

9. Communication and Monitoring

The final results must be communicated to stakeholders in an understandable manner.

Data scientists may use reports, dashboards, charts, presentations, and data visualizations. After deployment, the model should also be monitored because real-world data can change over time.

Example of the Complete Data Science Process

Step	Activity
<i>Problem Definition</i>	<i>Predict whether a customer will purchase</i>
<i>Data Collection</i>	<i>Collect customer and transaction data</i>
<i>Data Cleaning</i>	<i>Remove duplicates and handle missing values</i>
<i>EDA</i>	<i>Analyze purchasing patterns</i>
<i>Feature Engineering</i>	<i>Create useful customer features</i>
<i>Model Building</i>	<i>Train a classification model</i>
<i>Evaluation</i>	<i>Measure accuracy, precision, recall, etc.</i>
<i>Deployment</i>	<i>Integrate model into shopping website</i>
<i>Monitoring</i>	<i>Continuously check model performance</i>

Conclusion: *The Data Science Process provides a structured approach to solving real-world problems using data. Each stage, from problem definition to deployment and monitoring, is important for obtaining reliable and useful results.*

2. DATA WAREHOUSE

Introduction

A Data Warehouse is a centralized repository used to store large amounts of historical and integrated data collected from multiple sources.

It is mainly designed for reporting, analysis, and decision-making, rather than day-to-day transaction processing.

For example, a large retail company may collect data from sales systems, customer databases, inventory systems, online stores, and payment systems. This data can be integrated into a central data warehouse for analysis.

Definition

A Data Warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data used to support management decision-making.

Characteristics of a Data Warehouse

1. Subject-Oriented

A data warehouse is organized around major business subjects rather than individual applications. Common subjects include customers, sales, products, employees, and suppliers. Example: A retail data warehouse may focus on sales analysis rather than individual sales transactions.

2. Integrated

Data is collected from different sources and converted into a common format. For example, one system may store gender as M / F while another stores Male / Female. The data warehouse standardizes these values.

3. Time-Variant

A data warehouse stores historical data over a long period. Example: 2023 - Rs.20 lakh, 2024 - Rs.27 lakh, 2025 - Rs.35 lakh, and 2026 - Rs.42 lakh. This allows organizations to analyze trends over time.

4. Non-Volatile

Once data is stored in a data warehouse, it is generally not frequently modified or deleted like operational transaction data. Data is primarily used for reading and analysis. Therefore, the data remains relatively stable.

Architecture of a Data Warehouse

Basic flow: Data Sources -> ETL -> Data Warehouse -> OLAP -> Reports / Analysis

1. Data Sources

Data comes from various sources such as relational databases, ERP systems, CRM systems, Excel files, websites, APIs, and external databases.

2. ETL

ETL stands for Extract, Transform, Load.

Extract: Data is extracted from different sources.

Transform: The extracted data is cleaned and converted into a suitable format. Transformation may include data cleaning, removing duplicates, formatting, data integration, and aggregation.

Load: The transformed data is loaded into the data warehouse.

Example of ETL

Suppose a company has sales data in three different databases. **Extract:** Collect data from all three databases. **Transform:** Standardize product names, dates, prices, etc. **Load:** Store the cleaned data in the data warehouse.

3. Data Warehouse

The transformed data is stored in a centralized warehouse. It contains large amounts of historical data that can be used for analysis.

4. OLAP

OLAP stands for Online Analytical Processing. OLAP allows users to analyze data from multiple dimensions.

For example, sales can be analyzed by product, region, customer, year, and month.

OLAP Operations

- 1. Roll-up:** Moves from detailed data to summarized data. Example: City -> State -> Country.
- 2. Drill-down:** Moves from summarized data to more detailed data. Example: Year -> Quarter -> Month -> Day.
- 3. Slice:** Selects a particular dimension value. Example: Analyzing sales only for 2026.
- 4. Dice:** Selects data based on multiple conditions. Example: Analyzing sales for Chennai, Bangalore, Electronics products, and 2026.
- 5. Pivot:** Changes the way data is viewed by rotating dimensions. For example, switching rows and columns in an analytical report.

Advantages of Data Warehouse

1. Improves decision-making
2. Stores historical data
3. Integrates data from multiple sources
4. Supports business intelligence
5. Improves reporting and analysis
6. Helps identify trends and patterns
7. Reduces the workload on operational databases
8. Provides a single source of information

Data Warehouse vs Operational Database

Data Warehouse	Operational Database
<i>Used for analysis</i>	<i>Used for transactions</i>
<i>Stores historical data</i>	<i>Mainly stores current operational data</i>
<i>Supports OLAP</i>	<i>Supports OLTP</i>
<i>Complex queries</i>	<i>Frequent simple transactions</i>
<i>Mostly read-oriented</i>	<i>Read/write oriented</i>
<i>Used by analysts and managers</i>	<i>Used by operational users</i>
<i>Optimized for analytical queries</i>	<i>Optimized for transaction processing</i>

Data Warehouse in Data Science

A data warehouse is important in data science because it provides clean, integrated, historical data for analysis and modeling.

Operational Systems -> Data Warehouse -> Data Analysis -> Machine Learning -> Business Decision

A data scientist can use warehouse data to identify customer behavior, predict sales, analyze business performance, detect trends, build machine-learning models, and generate business reports.

Important Points to Remember

Data Science Process:

Problem -> Collect -> Clean -> Explore -> Engineer -> Model -> Evaluate -> Deploy -> Monitor

Data Warehouse characteristics:

S - Subject-oriented

I - Integrated

T - Time-variant

N - Non-volatile

ETL: *Extract -> Transform -> Load*

OLAP: *Online Analytical Processing*

Main OLAP operations: *Roll-up, Drill-down, Slice, Dice, Pivot*

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